

PREDICTING SEGMENT RAIL DISRUPTIONS USING IMBALANCE- AWARE MACHINE LEARNING

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Rail disruption is a rare-event prediction problem

Disruptions cascade

- Minor delays on one segment ripple into network-wide knock-on effects, resulting in missed connections, overcrowding, lost time and trust in the service.

But disruptions are rare

- On-time movements dominate the dataset. A naive model can score high on accuracy by ignoring disruptions entirely, which is useless for early warning.

Aim & Objectives

Aim: Build and evaluate an imbalance-aware early-warning prototype that predicts segment-level rail disruption risk and outputs calibrated risk scores for decision support.

Dataset & labelling

Public NJ Transit + Amtrak (NEC) data; segment = directional from→to; reproducible delay-threshold label

Imbalance handling

Class weighting as default; SMOTE applied train-only as a sensitivity check, never on test data

Leakage Safe Pipeline

Past-only features; calendar effects and rolling segment history shifted to avoid future info

Calibrated risk outputs

Probability calibration + tuned operating threshold; ranked segments grouped into risk tiers

Baseline & comparators

Logistic Regression (interpretable baseline) vs Random Forest vs XGBoost under identical splits

Reproducible evaluation

Time-aware splits, PR-AUC / precision / recall / F1 over accuracy,

Two challenges shape every design decision

Class imbalance

Disruptions are the minority class

- Accuracy is misleading ,predicting "no disruption" looks strong but catches nothing
- Stricter thresholds (≥ 10 min) shrink the positive class further and increase imbalance substantially.
- Evaluation prioritises PR-AUC, precision, recall and F1 not accuracy

Temporal leakage

Time-ordered data breaks random splits

- Random shuffling lets models learn from the future inflated metrics, no real-world value
- Splits must be chronological: train on earlier, validate intermediate, test on later
- Lag features must be shifted; SMOTE applied train-only, never to validation or test

End-to-end leakage-safe pipeline

Guardrails embedded in the pipeline

Time-aware splits

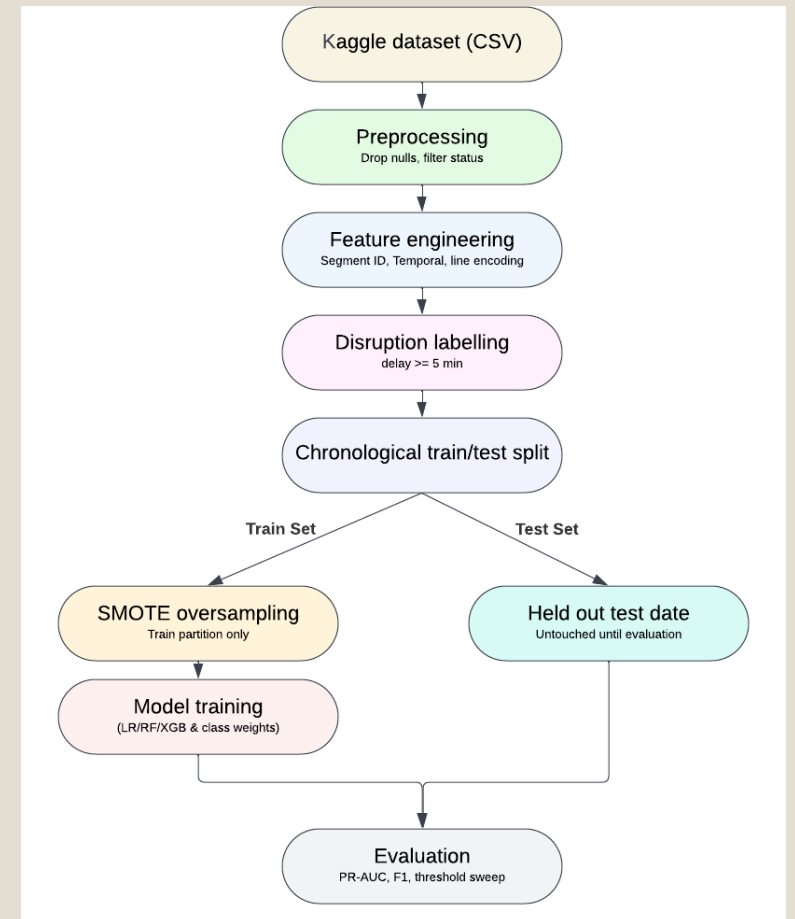
- Train on earlier periods, test on later so no random shuffling

Train-only features

- SMOTE never touches validation or test partitions

Past-only features

- Rolling segment history shifted to use only prior information



NJ Transit + Amtrak (NEC) - segment-level labelling

Why this dataset

Stop-level performance records with timestamps and delay measurements which are consistent enough for supervised learning under public-data constraints.

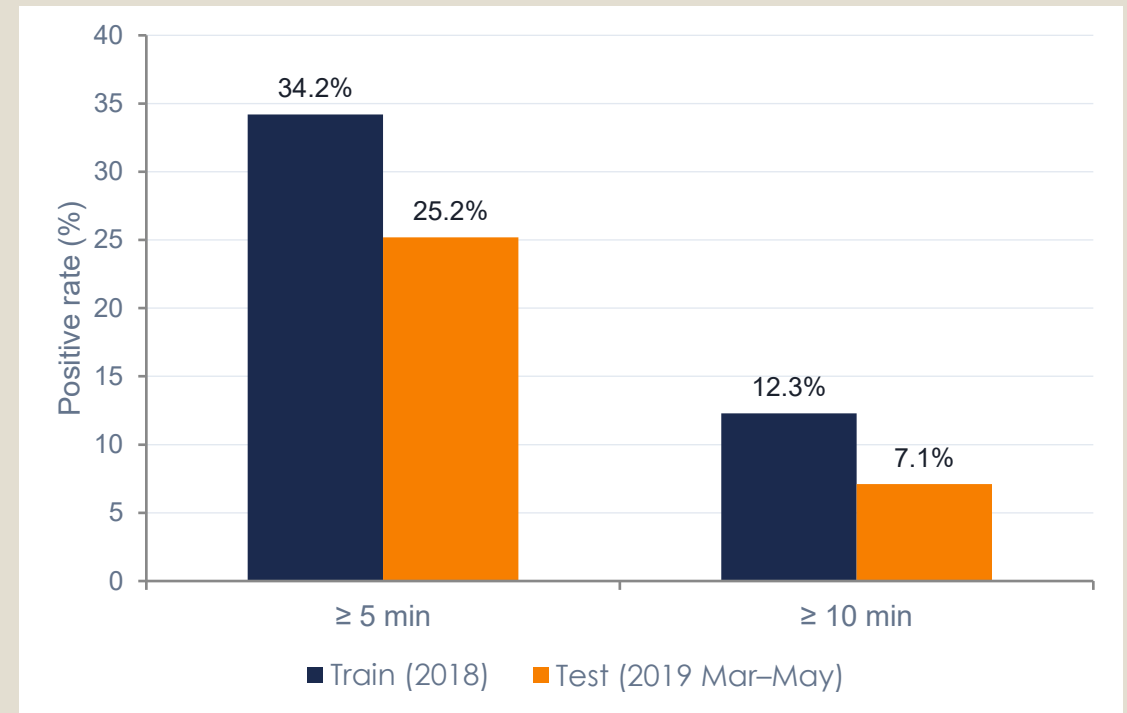
Why a delay threshold (not incident causes)

- Public datasets rarely provide stable, reliable incident-cause labels. A transparent ≥ 5 min delay rule is reproducible and lets the model focus on disruption risk and not causal diagnosis.

Why segment-level (not station-level)

- A station can be affected by upstream issues. A directional segment (from $\rightarrow$ to) localises risk to the specific link where disruption is observed.

Label distribution by threshold



Three iterative prototypes - risk-managed scope

P1+ P1.5

Feasibility baseline

1–3 months • LR baseline

- Single-month → 3-month window
- Established label, time split, PR-AUC
- Verified pipeline before scaling

Best PR-AUC Score: **0.29**

P2

Multi-model comparison

6 months • LR vs RF vs XGB

- Class weighting + SMOTE (train-only)
- Threshold sensitivity at 5 vs 10 min
- XGBoost + CW emerged strongest

◦ Best PR-AUC Score: **0.44**

P3

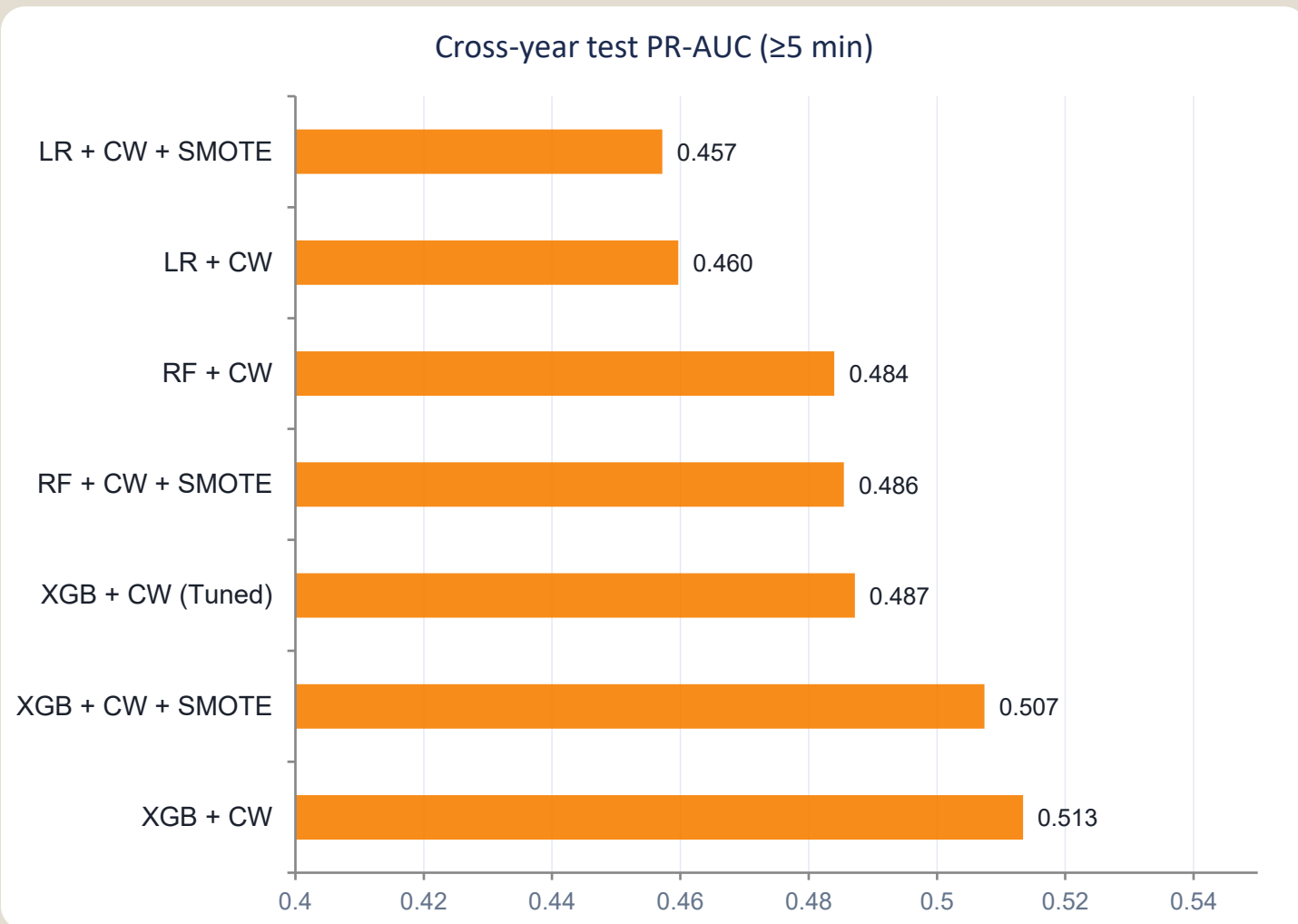
Cross-year + lag features

8 months • cross-year hold-out

- Leakage-safe 7-day rolling segment history
- Probability calibration + threshold tuning
- Ranked segments + risk tiers

◦ Best PR-AUC Score: **0.51**

XGBoost + class weighting wins on PR-AUC



Headline metrics (cross-year test)
Best model XGB + Class weights

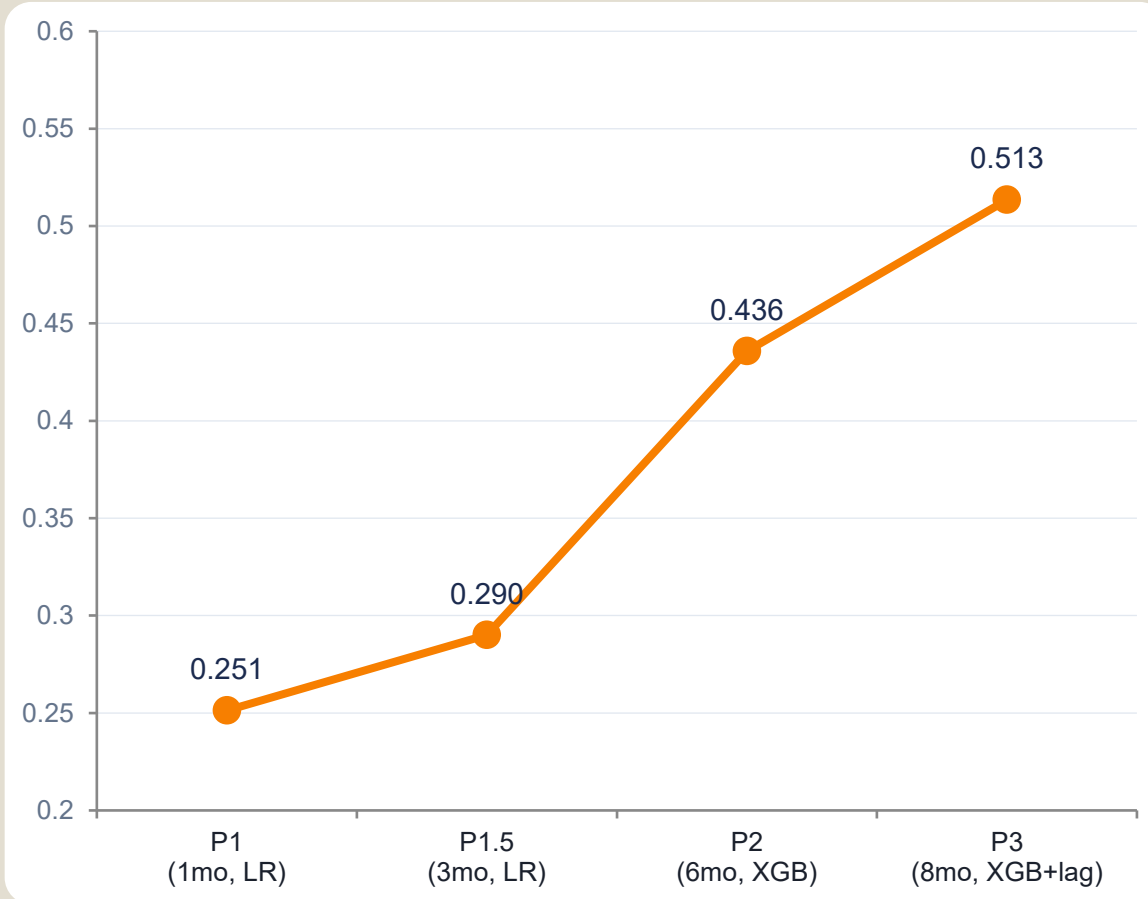
- PR-AUC **0.513**
- F1 0.509
- Recall 0.552
- Precision 0.473

Why not SMOTE?

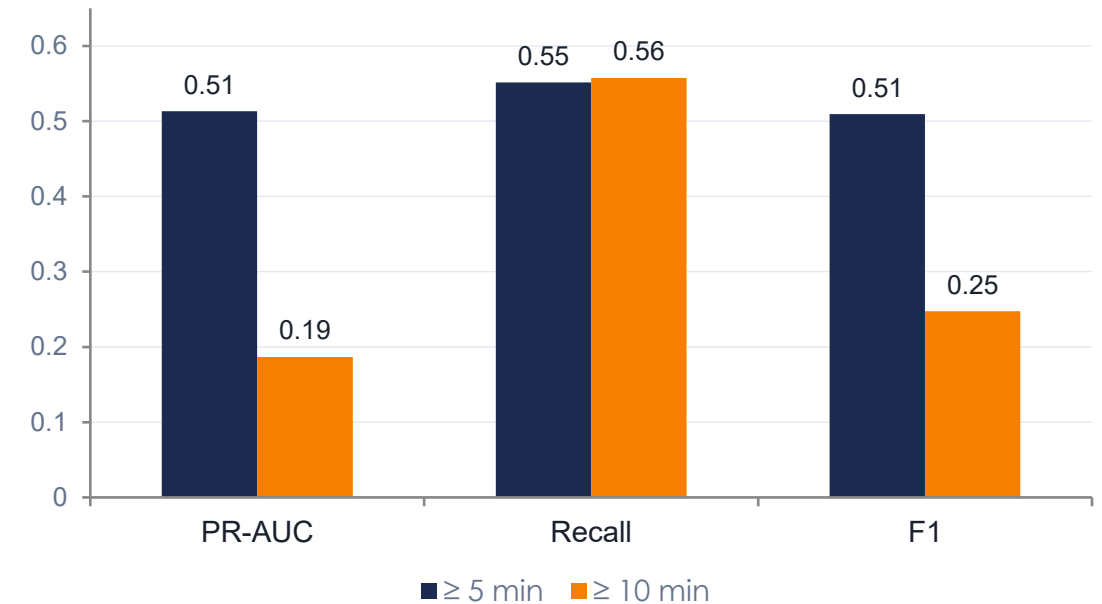
- Higher recall (0.71) but precision drops to 0.40 but too many false alarms for early warning.

Progression and severity sensitivity

PR-AUC across prototypes



Threshold sensitivity (P3, XGB + CW)



Stricter = harder

- Tightening to ≥ 10 min increases class imbalance substantially; PR-AUC and F1 drop sharply. ≥ 5 min stays the primary label; ≥ 10 min is reported as a sensitivity check.

Calibrated probabilities, tuned operating point

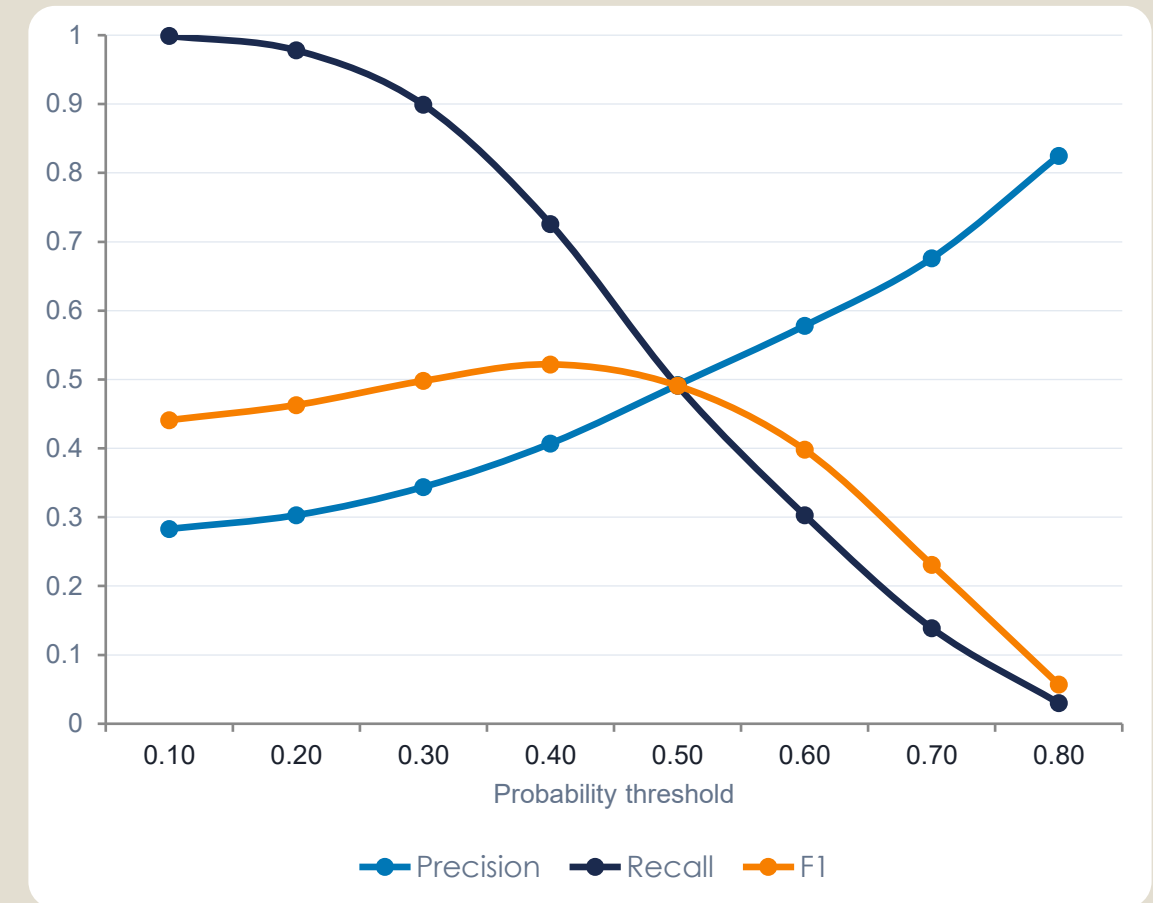
Why this matters

- A binary label tells you if a segment is risky. A calibrated probability tells you how risky it may be and lets operators rank segments and trade off recall vs precision against alert volume.

What was applied

- Platt and Isotonic calibration tested via reliability plots; predicted probabilities mapped close to observed disruption frequencies.
- Threshold sweep (0.10 → 0.85) selected 0.40 as the operating point and best F1 with non-trivial precision; appropriate where missed disruptions are costly.

Threshold sweep — Prototype 3



Ranked segments and risk tiers - actionable output

Risk tier distribution (cross-year test)



■ MINIMAL (38.3%) ■ LOW (33.1%)
■ MEDIUM (18.1%) ■ HIGH (10.5%)

Top 5 highest-risk segments

Rank	Segment	Line	Risk %	Actual	Obs	Tier
1	Linden → Newark Airport	Northeast Corrd	86.30%	100.00%	1	HIGH
2	Trenton → New Brunswick	Northeast Corrd	85.00%	98.40%	64	HIGH
3	Linden → Metropark	Northeast Corrd	82.90%	66.70%	3	HIGH
4	Linden → New Brunswick	Northeast Corrd	82.90%	100.00%	1	HIGH
5	Linden → Newark Penn Stn	No Jersey Coast	82.90%	100.00%	1	HIGH

Concentration of risk

4 of the top 5 risk segments sit on the Northeast Corridor, ranking surfaces real operational hotspots, not random noise.

Key findings

Boosted trees + class weighting was the strongest configuration

- XGB + CW achieved cross-year PR-AUC = 0.513 with stable F1 = 0.509. Robust to time shift.

SMOTE traded precision for recall but not the right fit here

- Class weighting kept false alarms manageable; SMOTE inflated recall but hurt PR-AUC.

Calibration turned a classifier into decision support

- Predicted probabilities aligned with observed segment rates → actionable ranking and tiers.

Lag features mattered more than model complexity

- Leakage-safe 7-day rolling segment history drove the P2 → P3 gain (0.44 → 0.51), not tuning.

Time-aware evaluation revealed the real performance

- Cross-year hold-out confirmed generalisation; random splits would have inflated metrics.

Threshold sensitivity exposed the severity trade-off

- ≥ 10 min tripled imbalance and halved PR-AUC to ≥ 5 min held as the primary, defensible label.

Limitations and where this goes next

LIMITATIONS

Threshold-based label

- ≥ 5 min is a proxy as it predicts disruption risk, not incident cause.

Single dataset family

- Trained on NJ Transit + Amtrak NEC; cross-network generalisation untested.

Sparse-segment variance

- Rare or new segments produce noisier risk scores than well-observed ones

Batch, not live

- Pipeline runs offline; no streaming ingestion or real-time scoring.

FUTURE WORK

Richer external features

- Weather summaries, planned engineering works, incident logs where available.

Cross-network validation

- Apply the pipeline to additional rail networks to test generalisation claims.

Graph-based propagation

- Model how disruptions cascade between connected segments using GNNs

Streaming early-warning tool

- Live ingestion + rolling lag updates → near real-time segment risk scores.

Thank you!

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<https://github.com/SamuelEdozie/rail-segment-disruption-prediction>

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